Distributed Quantum Computing

Massimo Equi

Aalto Univeristy

15 June 2026

Interview for IQM

About Me

Three main research topics

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String Matching
(for graphs)

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Quantum Computing

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Distributed Computing

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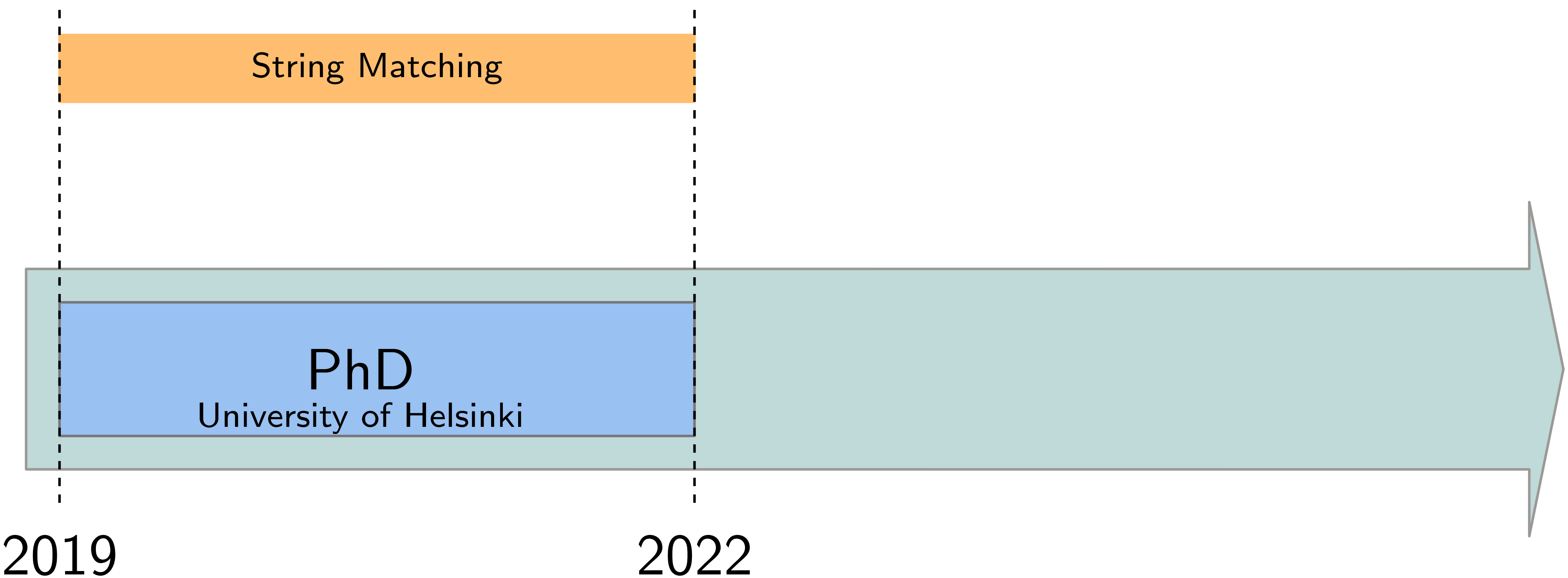
- Distributed Computing



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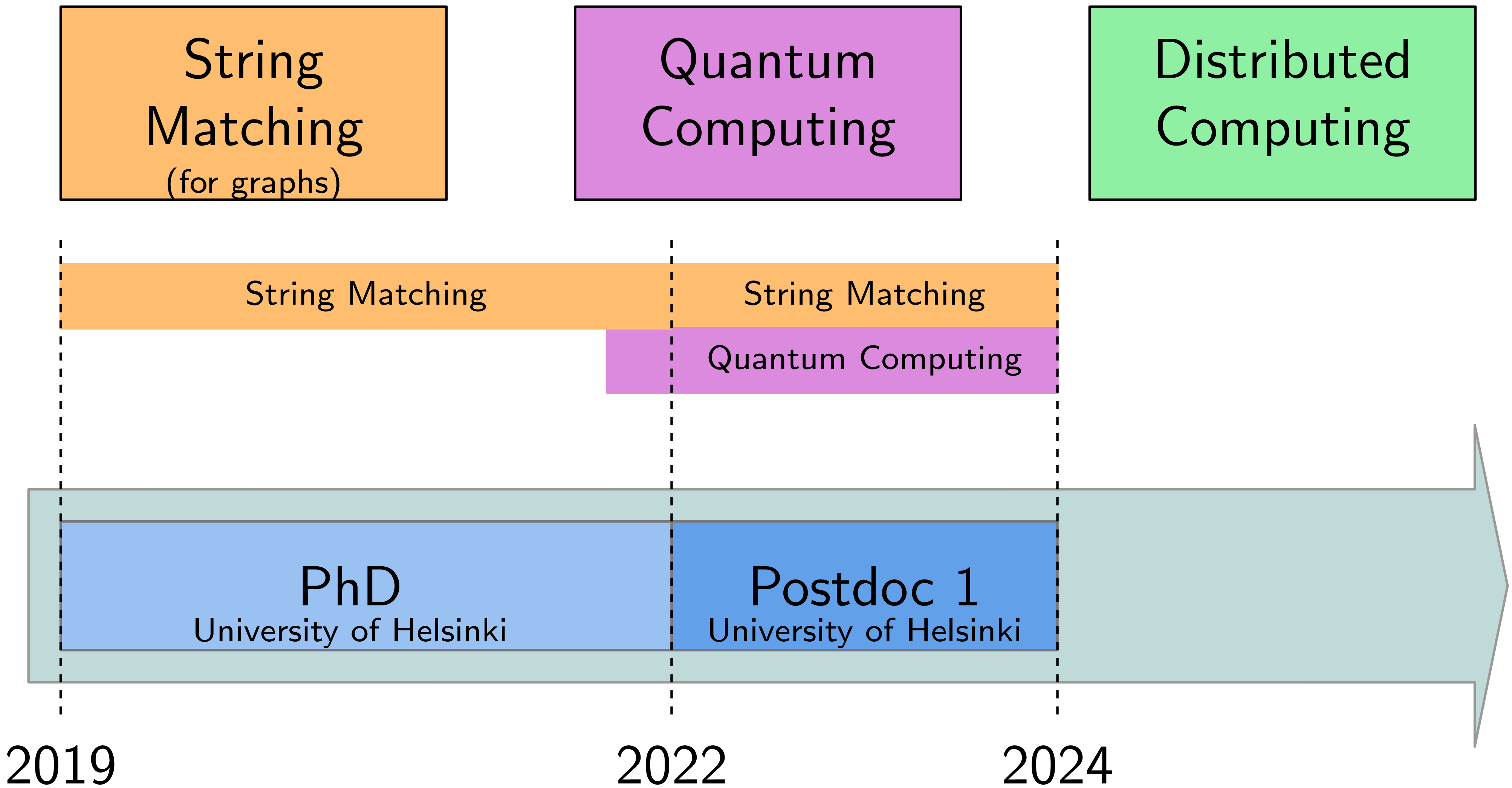

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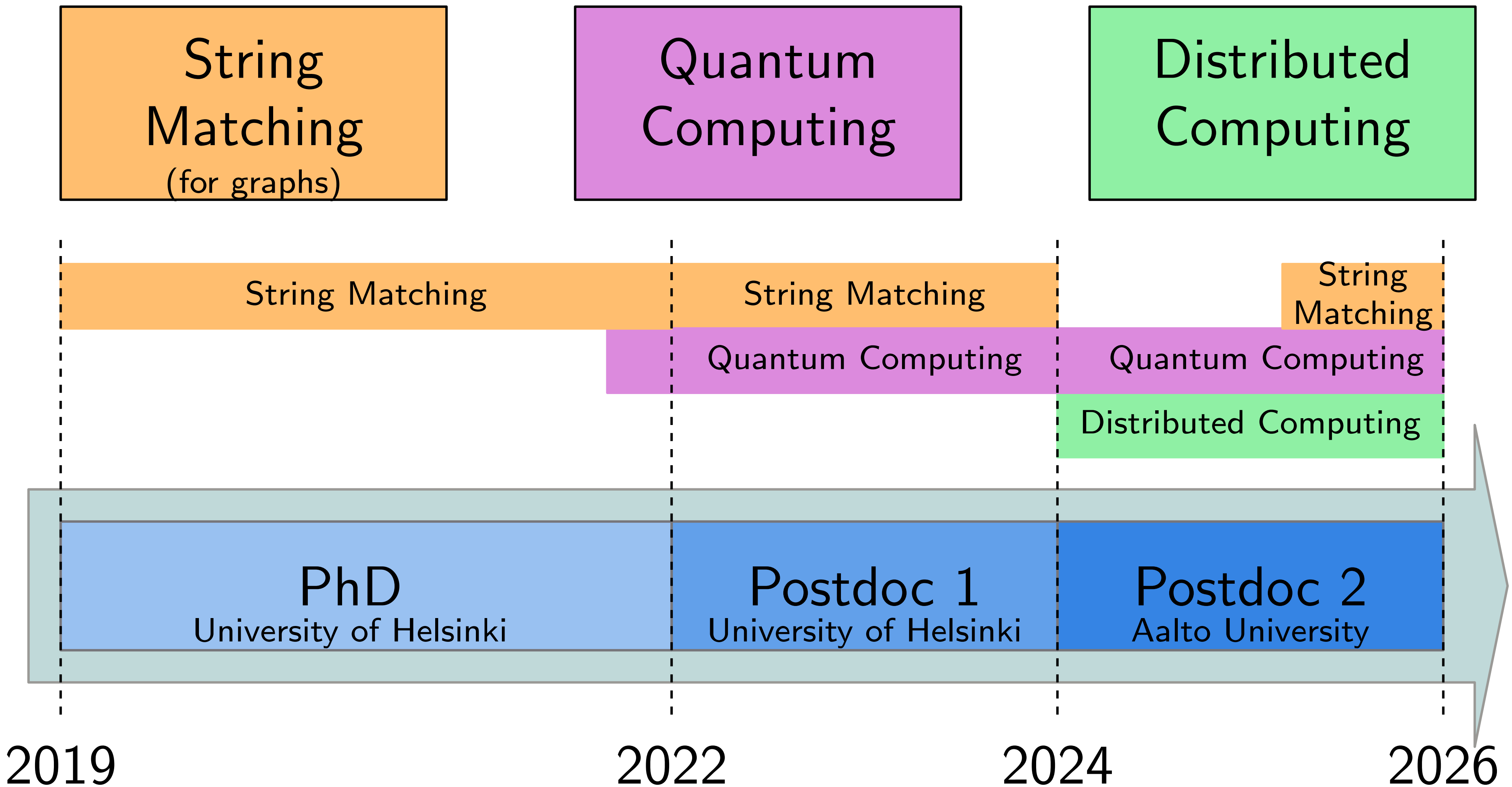
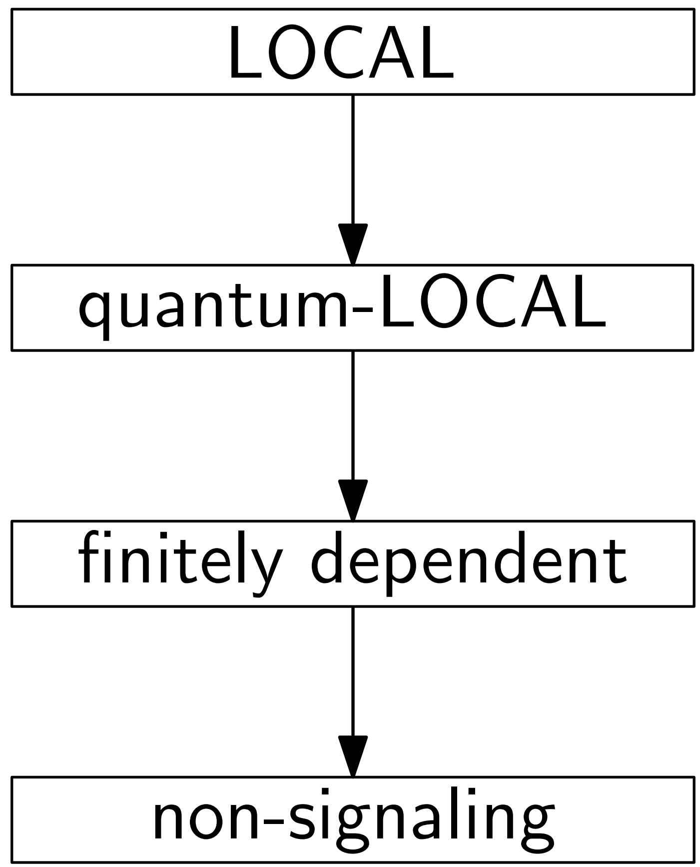
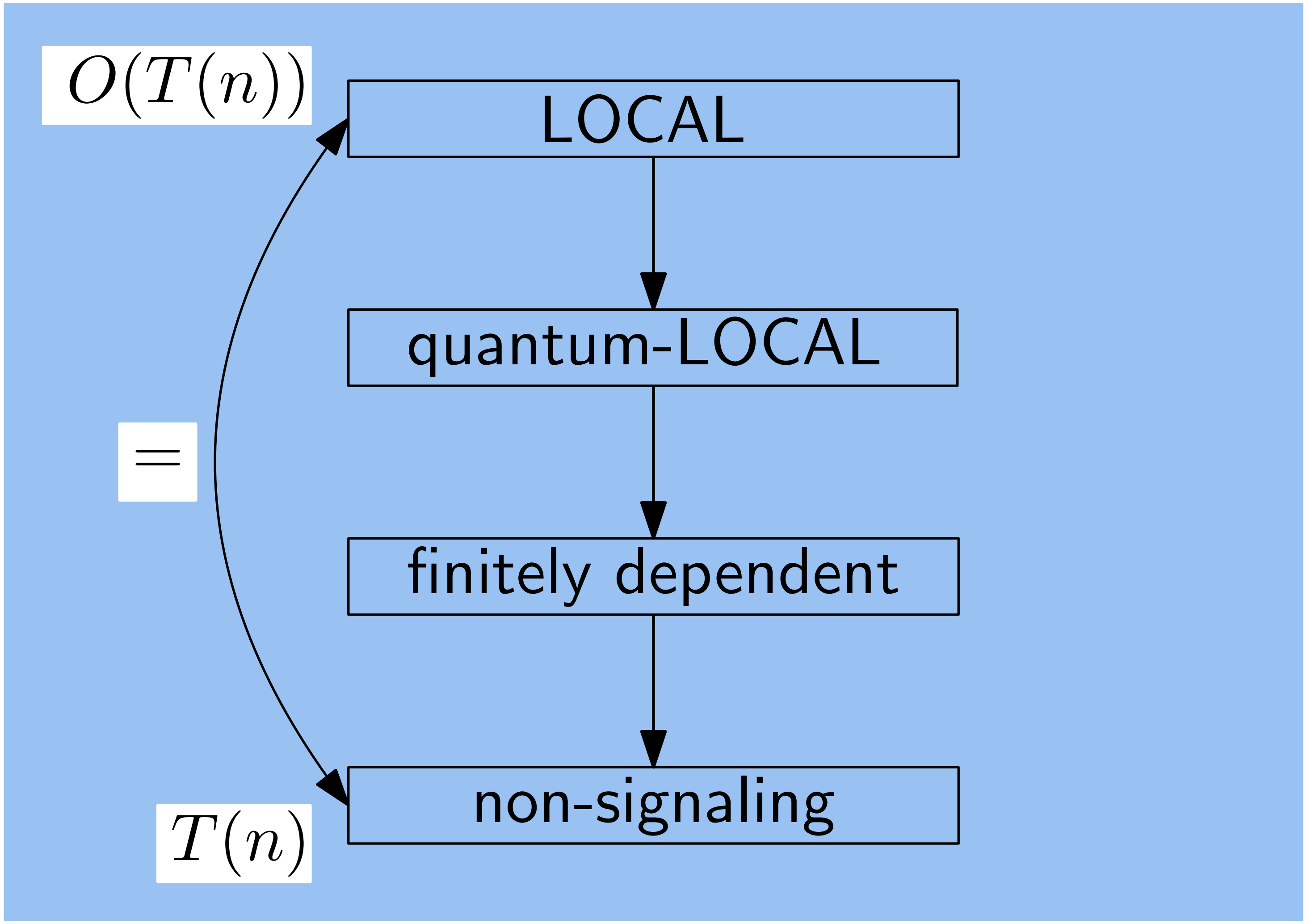


Distributed Hierarchy



Distributed Hierarchy

For Linear Programs (LPs)



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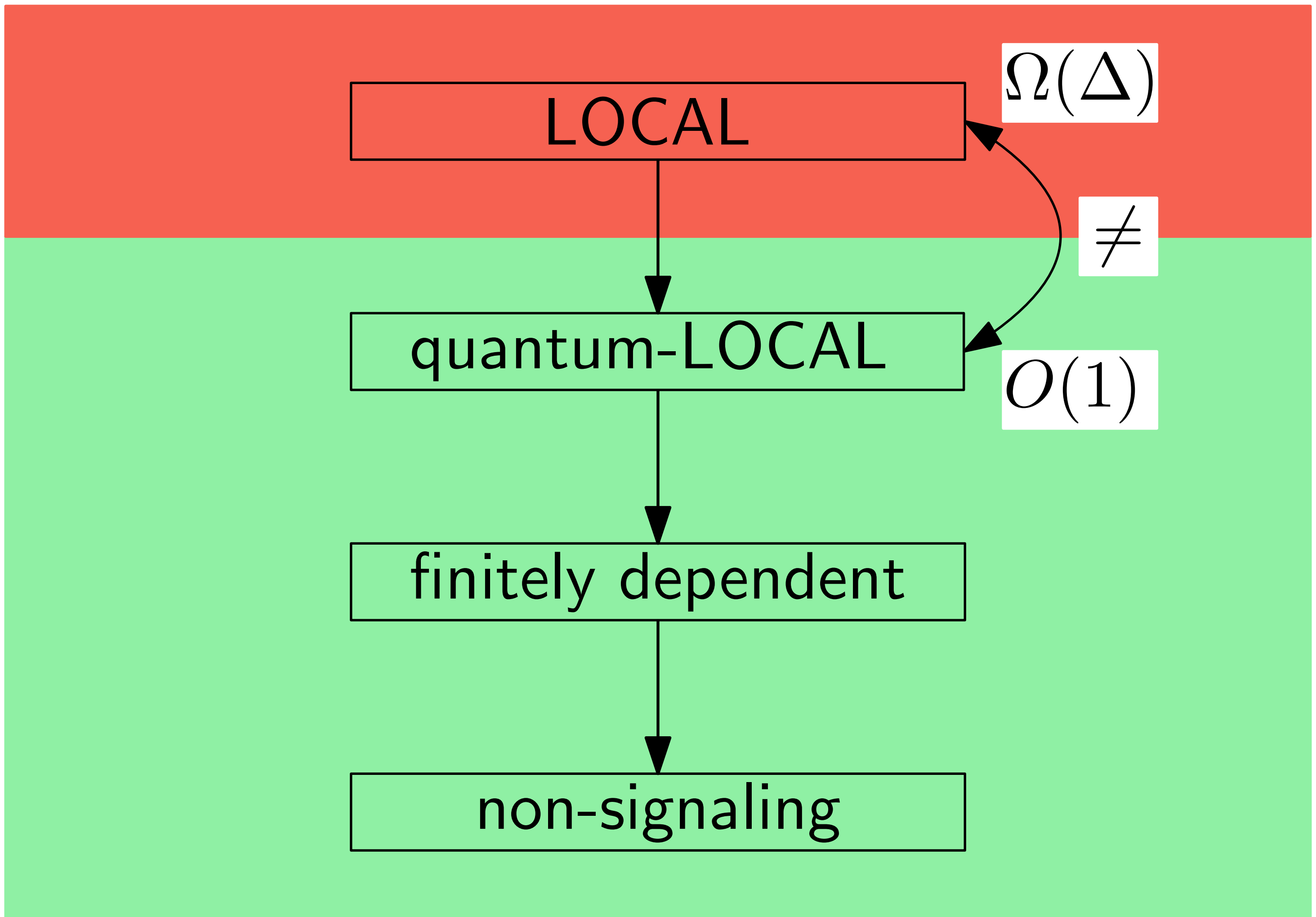
For Locally Checkable Labelling Problems (LCLs)



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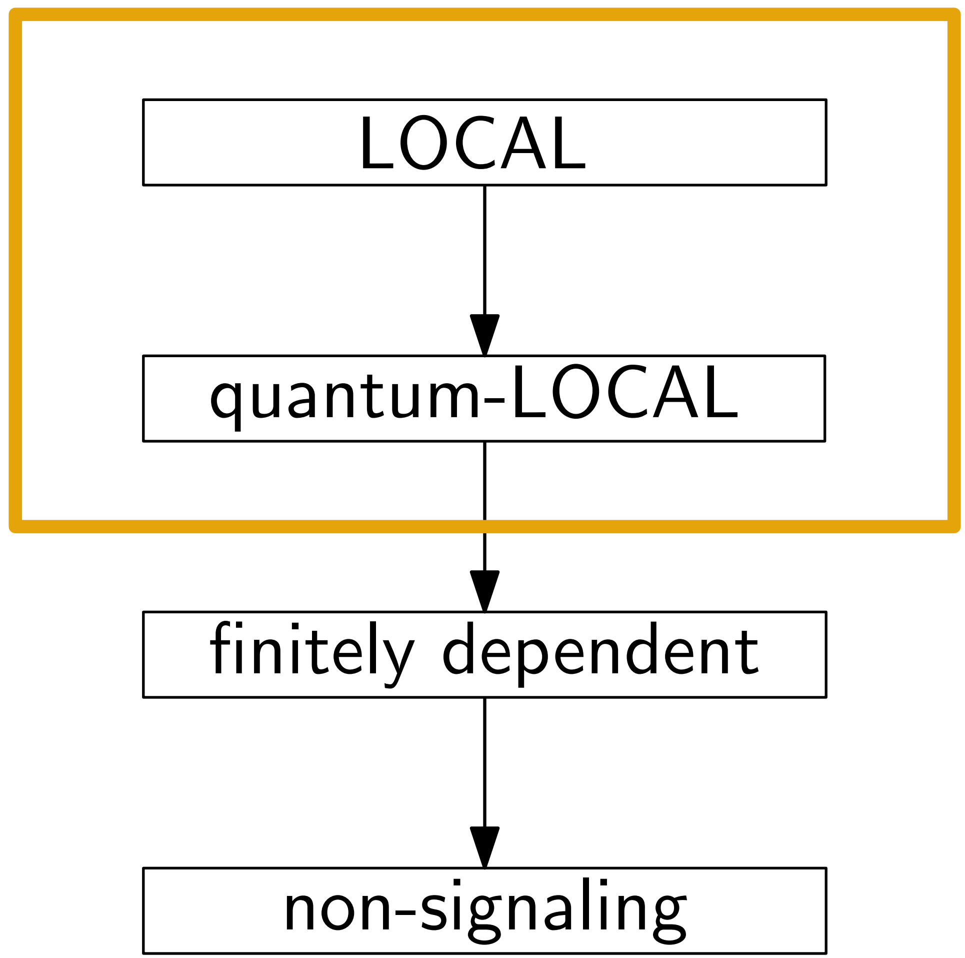
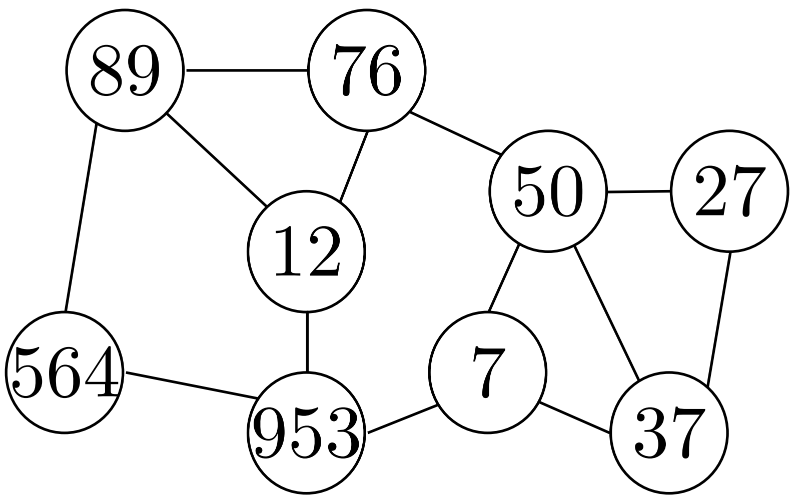


LOCAL and quantum-LOCAL



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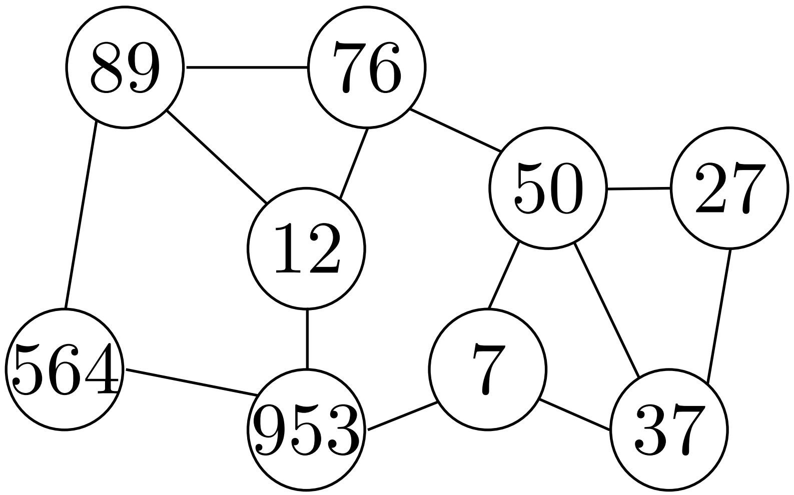

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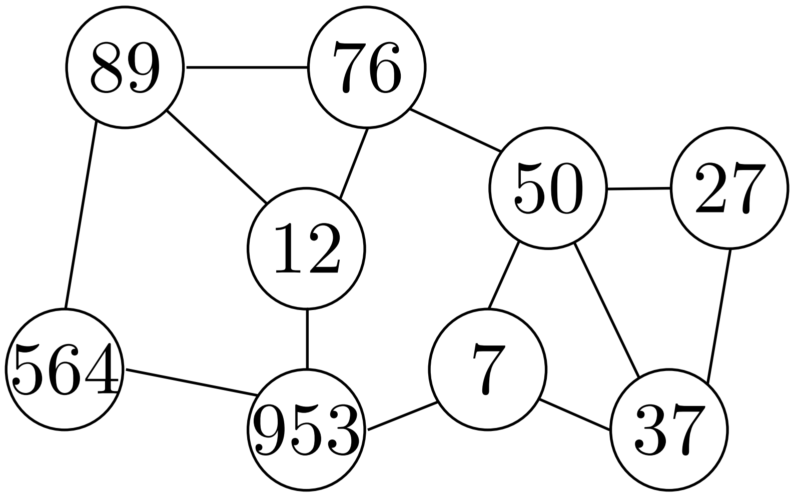

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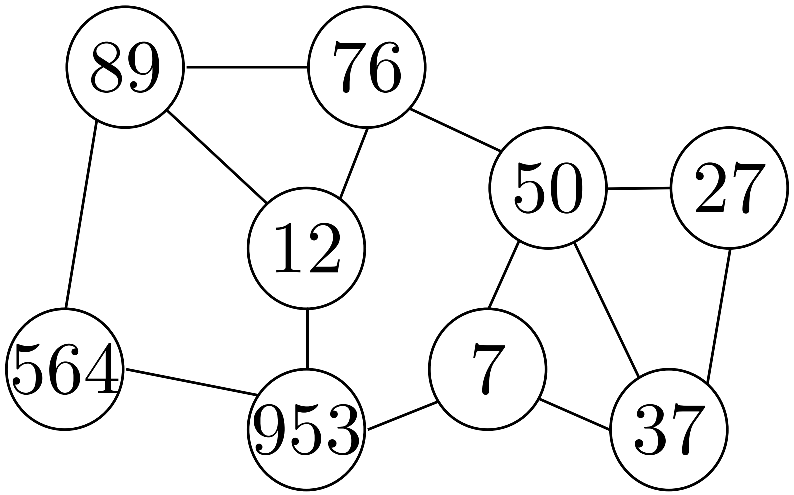

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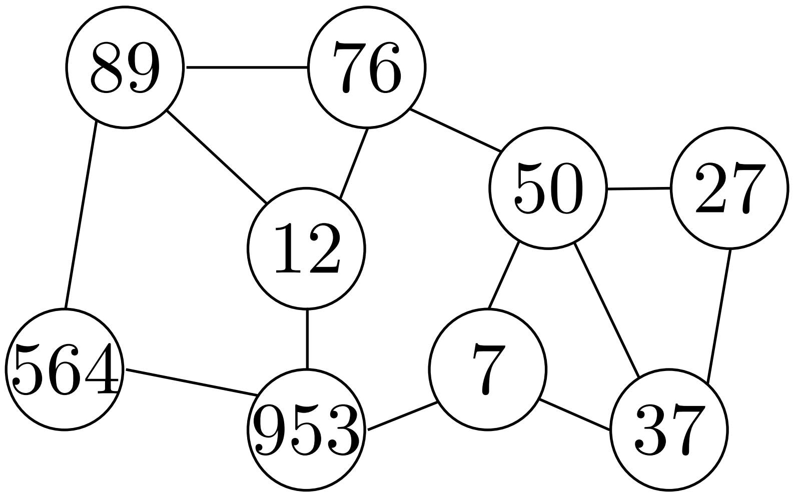

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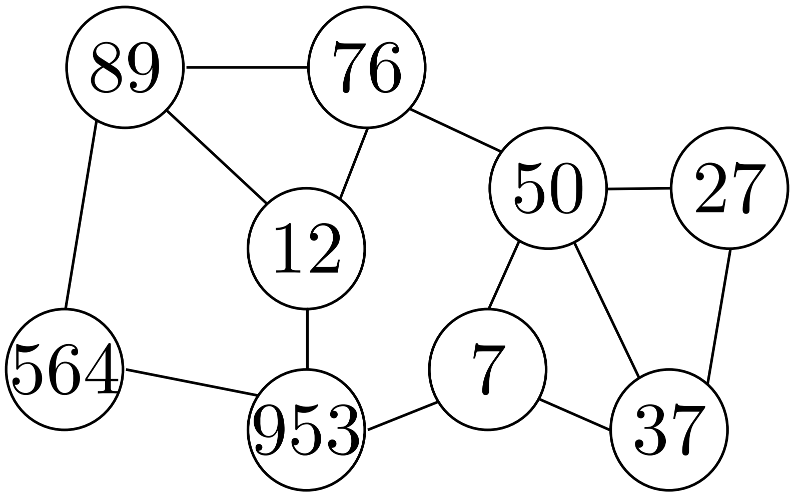

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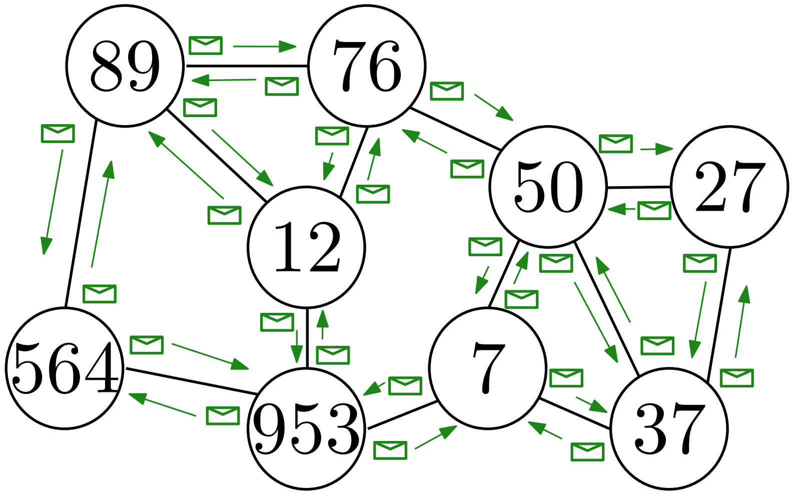

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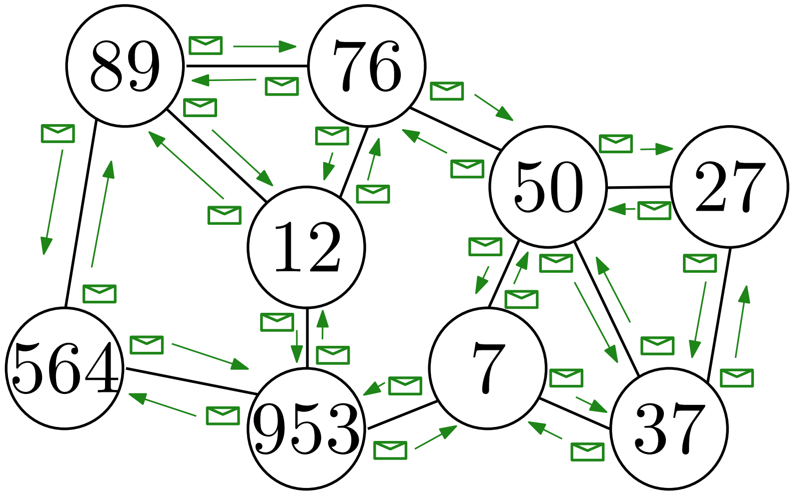

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Number of communication rounds as a function of n

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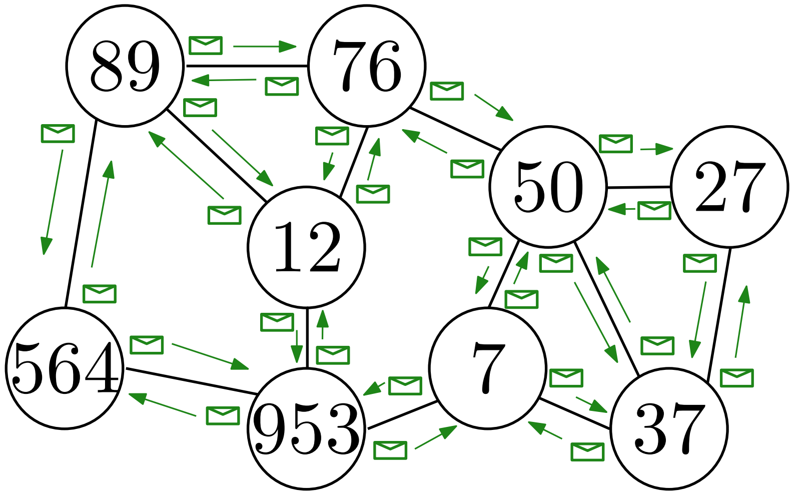

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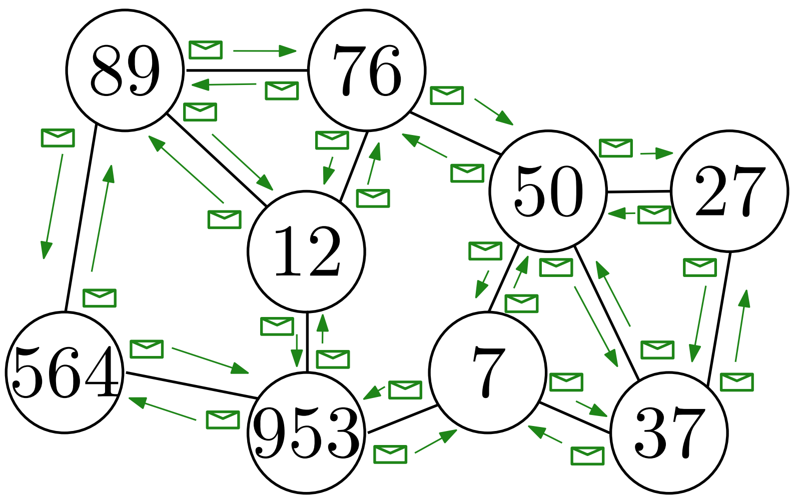

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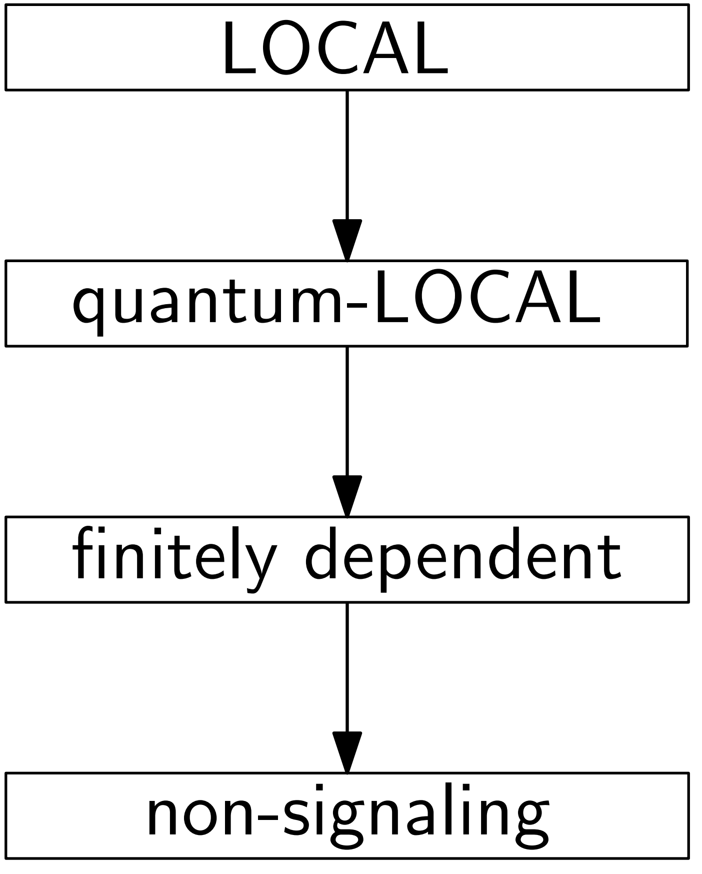
quantum-LOCAL

Nodes hold, send and receive **qubits**

IDs are not needed

"send qubits" means applying SWAP gates

Distributed Hierarchy



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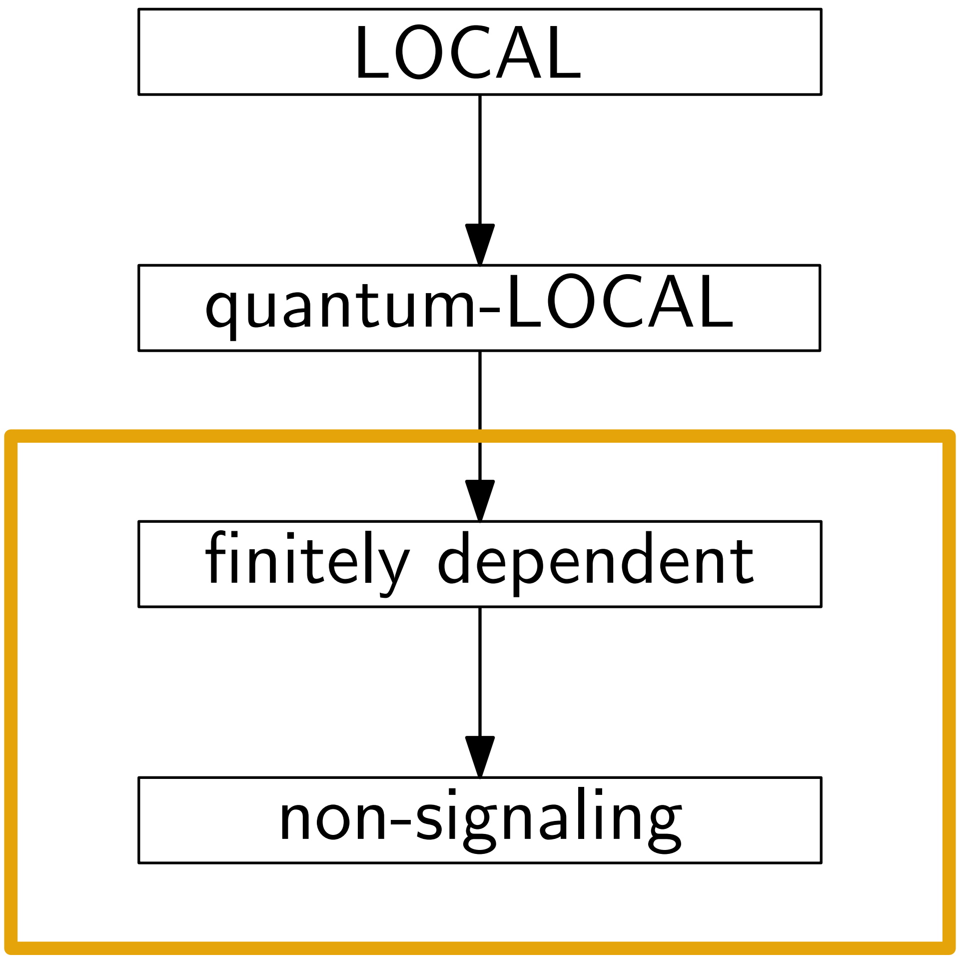
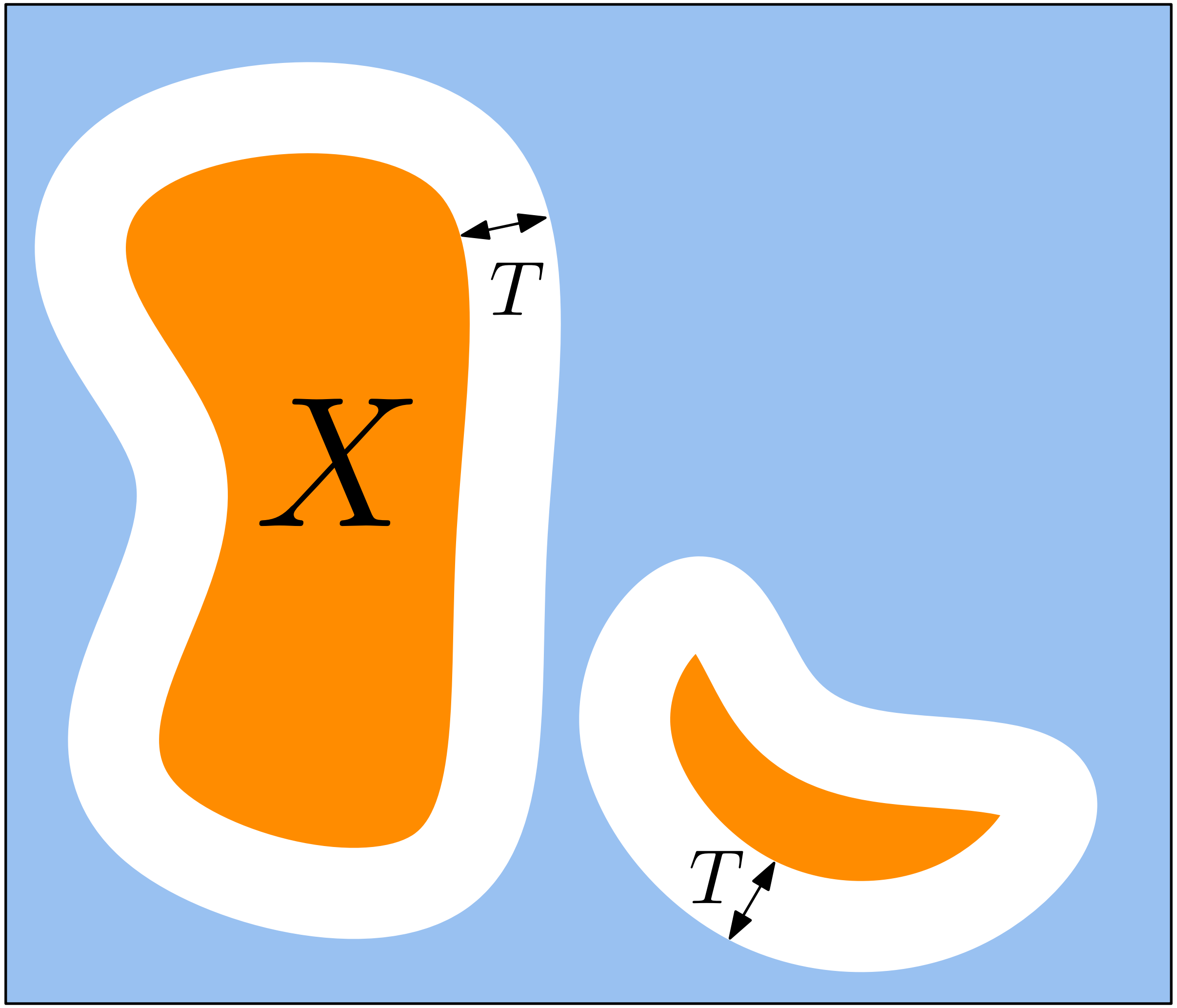


Non-signaling distributions

For any subset of nodes X, changes at distance $> T$ **do not affect** the output distribution of X



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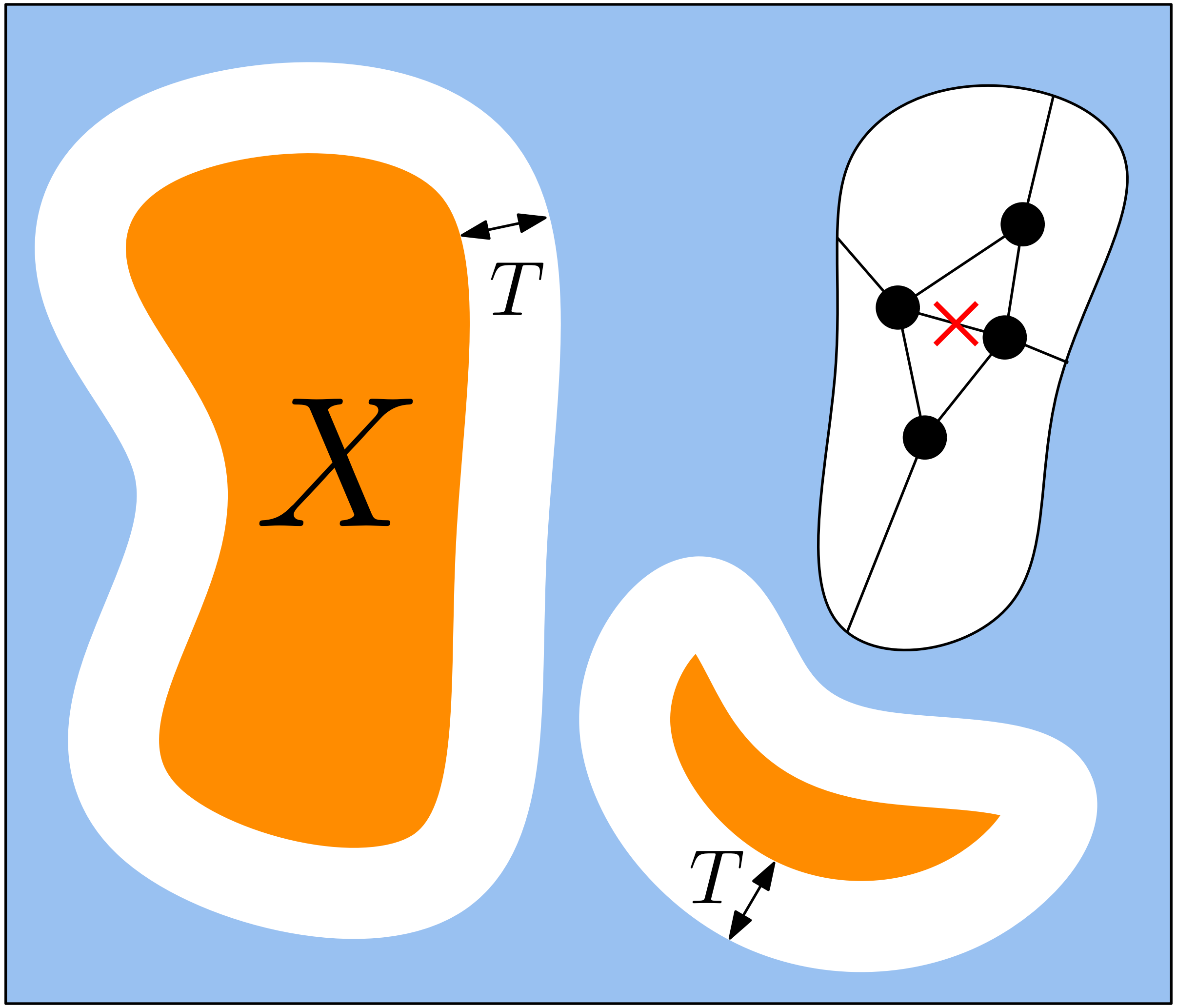

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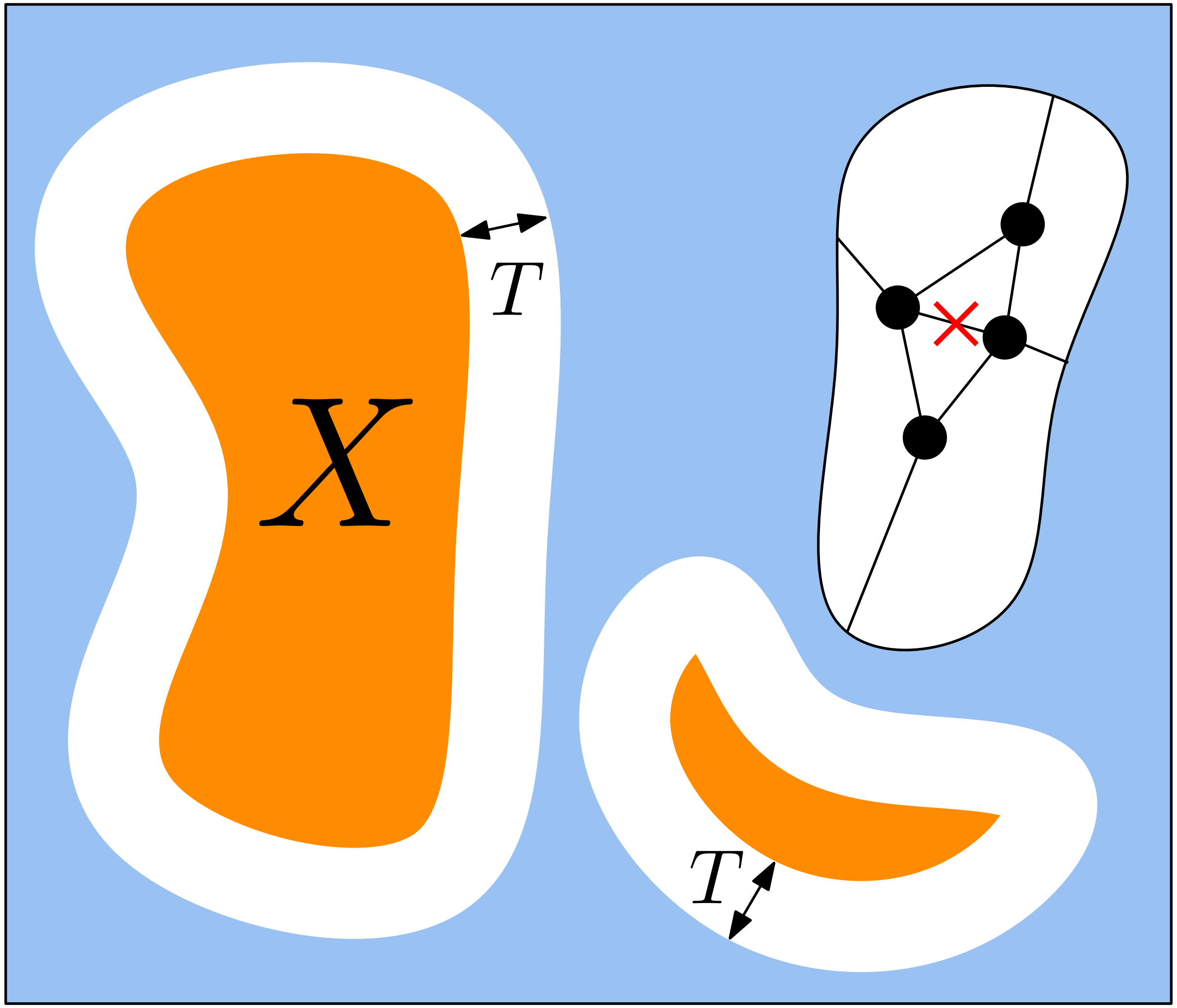

Finetely dependent distributions add restrictions not considered today

Results for Linear Programs (LPs)

Reminder:

- **Independent Set (IS)**:
  a subset of nodes $I \subseteq V$ such that $\forall v, u \in I : (v, u) \notin E$
- **Maximal Independent Set (MIS)**:
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Example of Distributed LP: **fractional maximum independent set**

$$\begin{aligned} \text{maximize} \sum_{v \in V} x_v & & \\ \text{subject to} \sum_{(v,u) \in E} x_v + x_u \leq 1 & & \forall v \in V \\ 0 \leq x_v \leq 1 & & \forall v \in V \end{aligned}$$

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When $x_v \in \{0, 1\} \quad \forall v \in V$, the solution is a **maximum independent set**, i.e. an MIS of largest cardinality.

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For LPs, LOCAL can simulate any non-signaling distribution

Example for fractional MIS

Let $T(n)$ be the locality of the non-sign. distribution

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Key facts

- By linearity of $\mathbb{E}$,

$$\sum_{v \in V} \mathbb{E}[X_v] = \mathbb{E}\left[\sum_{v \in V} X_v\right] = \mathbb{E}[X]$$

- By monotonicity of $\mathbb{E}$, if X is an α-approx. then $\mathbb{E}[X]$ is an α-approx.

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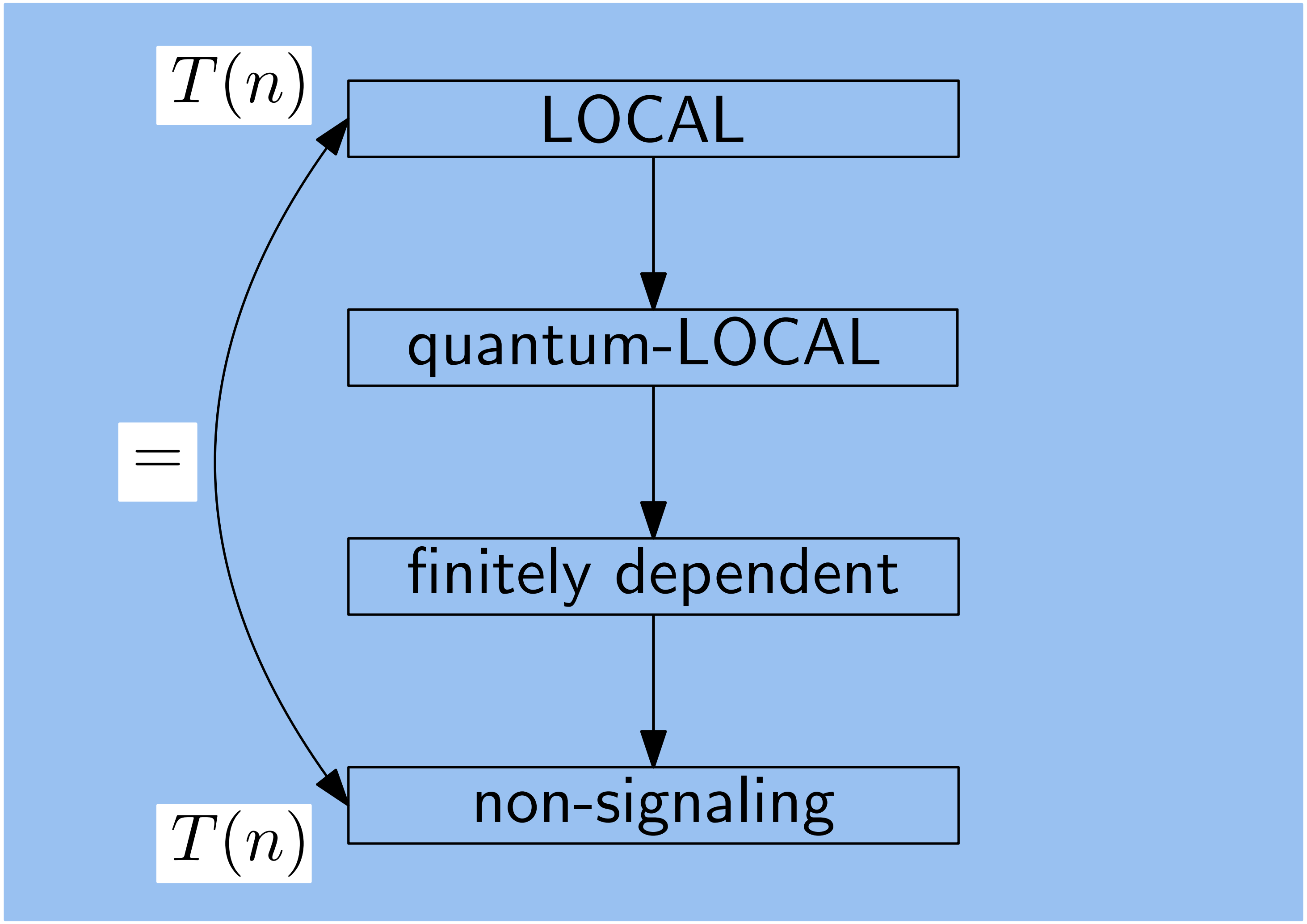

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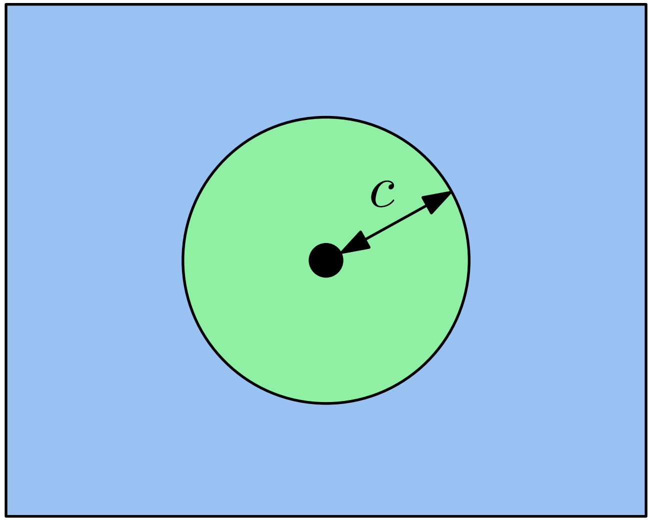
LCLs: explore up to radius c to check a solution



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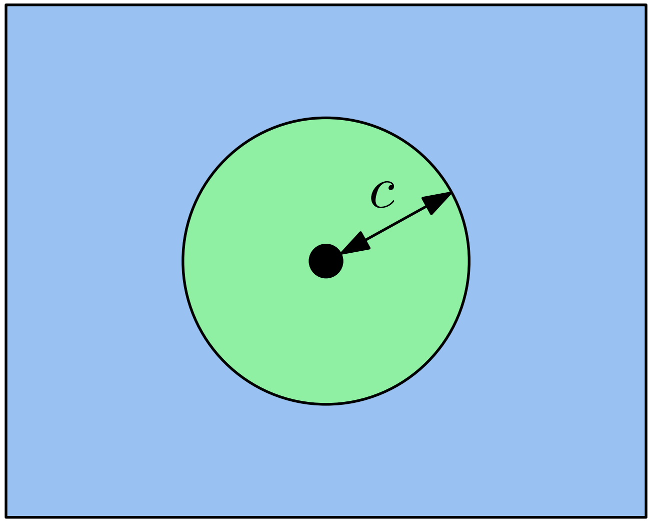

Examples

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- Coloring
- Maximal Matching
- iterated-GHZ

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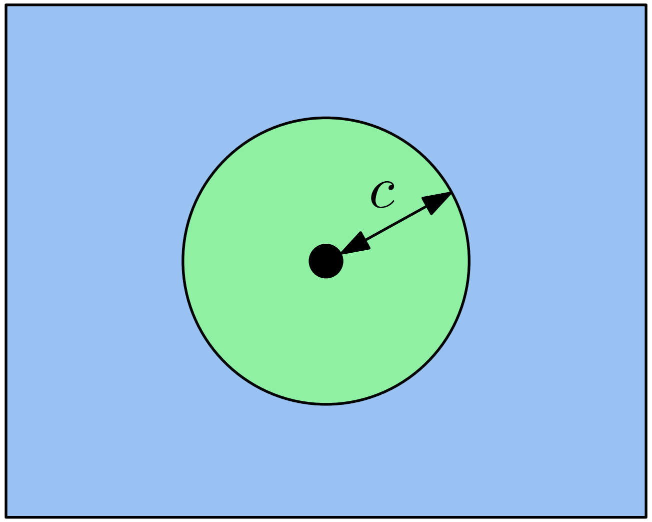

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Three players A, B and C, and one referee

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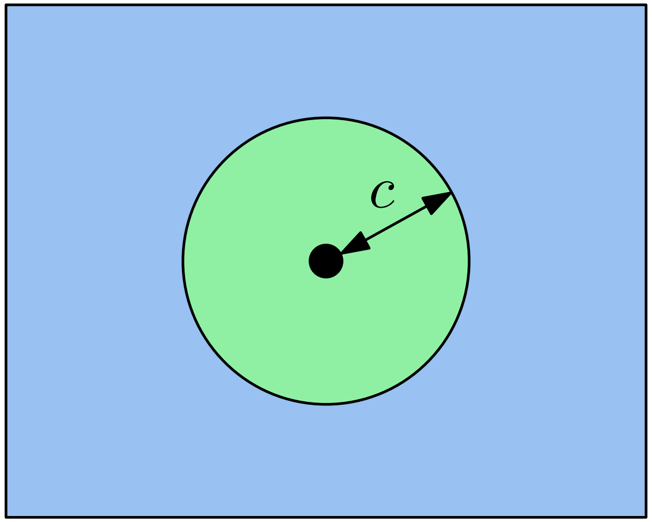

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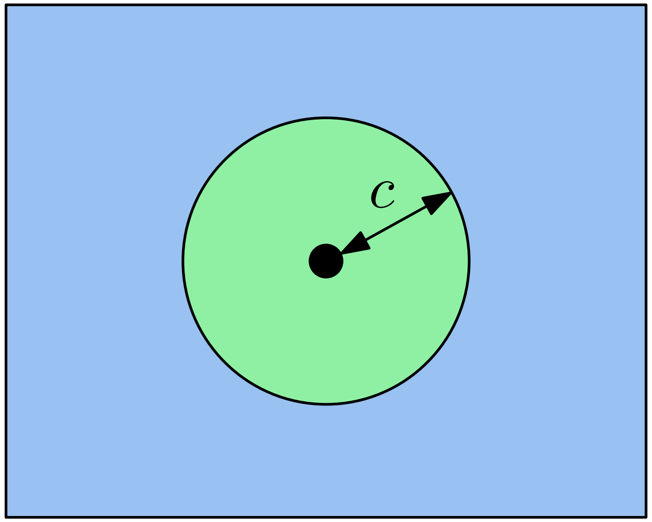

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Goal: **without communicating after the input is given**, output bits such that

$$a \oplus b \oplus c = x \vee y \vee z$$

knowing that $x \vee y \vee z \in \{(0,0,0),(1,1,0),(1,0,1),(0,1,1)\}$

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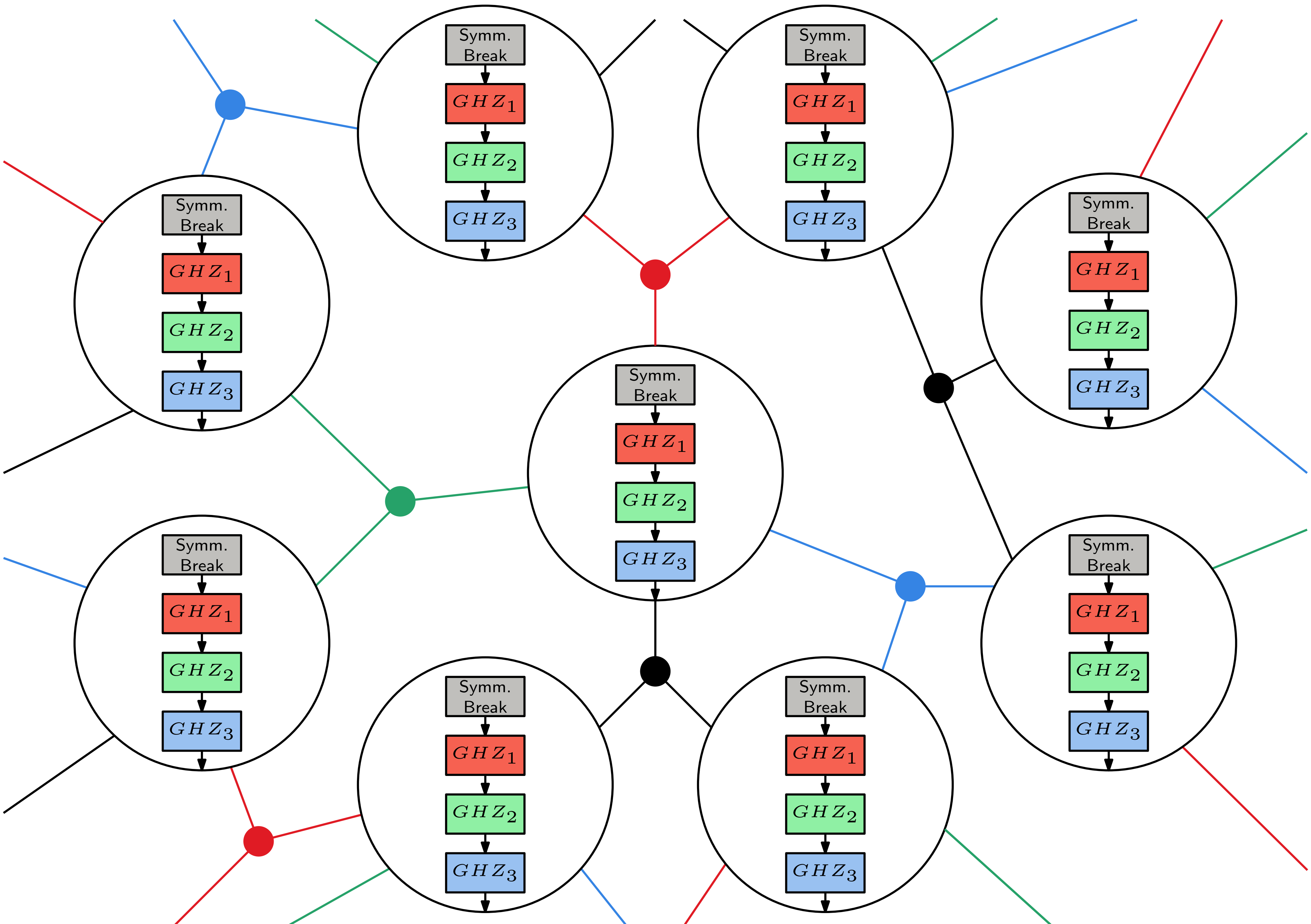

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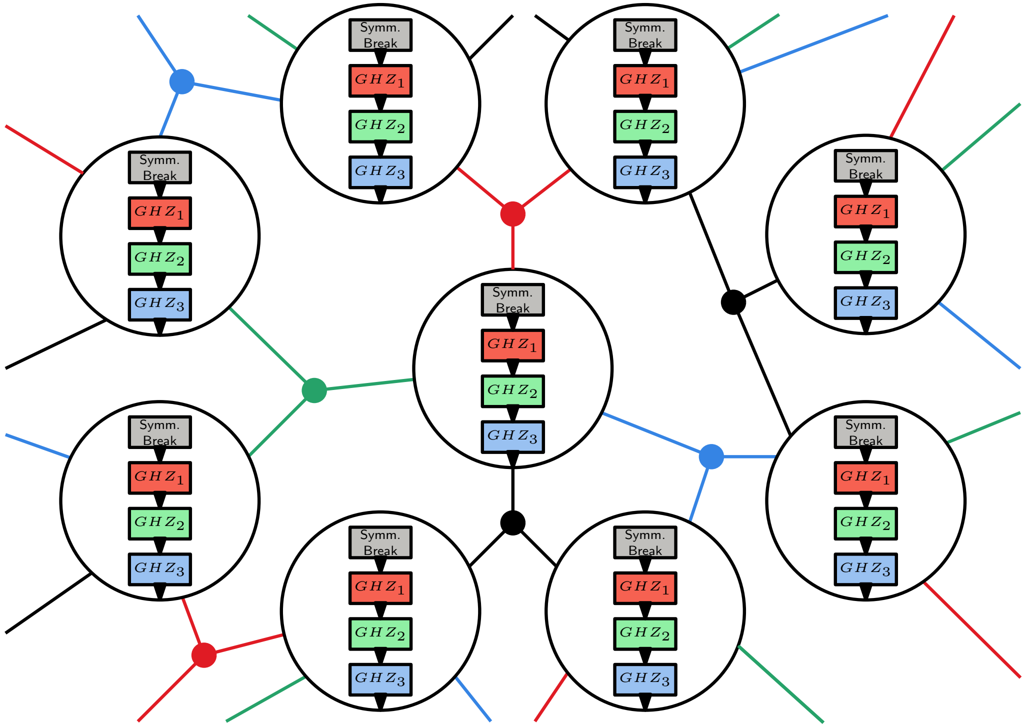
Round complexity of iterated-GHZ



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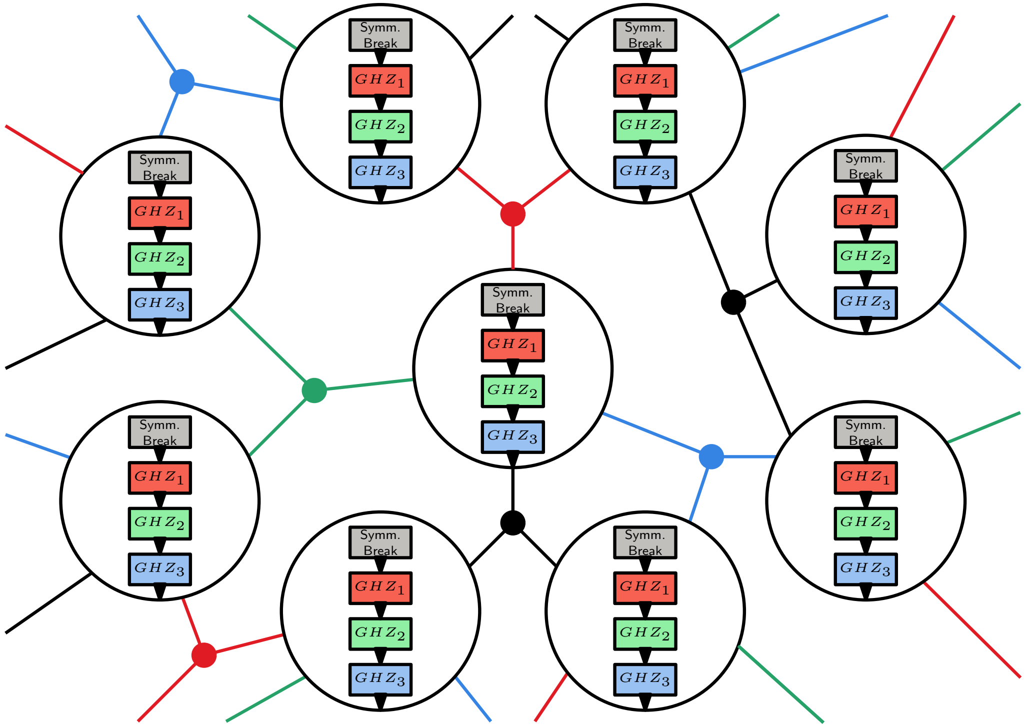

quantum-LOCAL: $O(1)$ rounds

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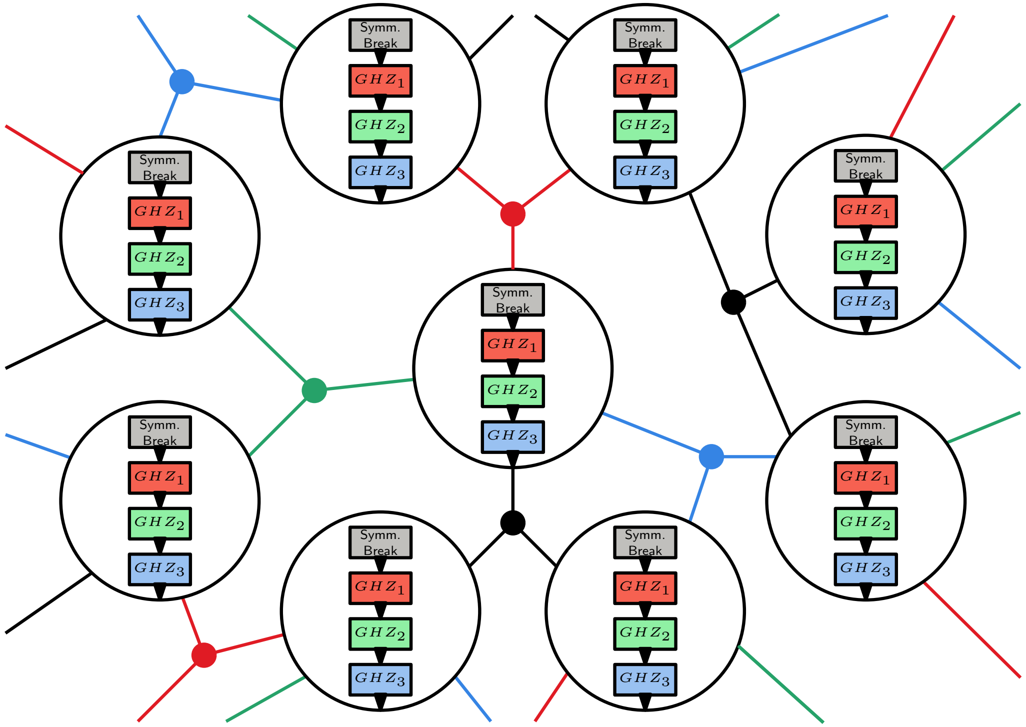

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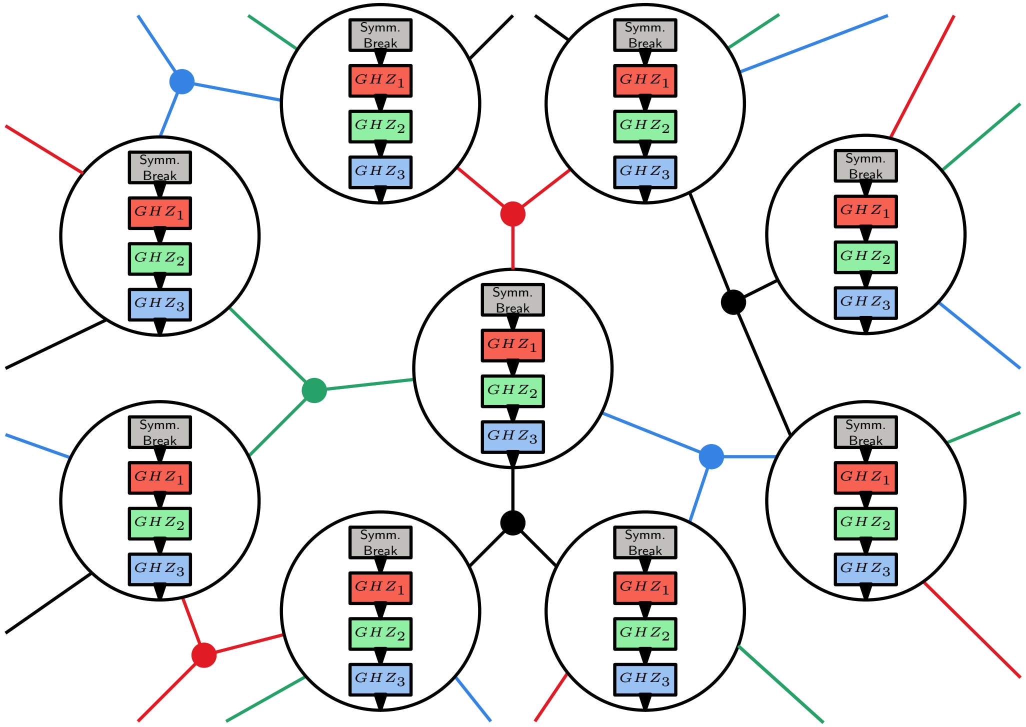

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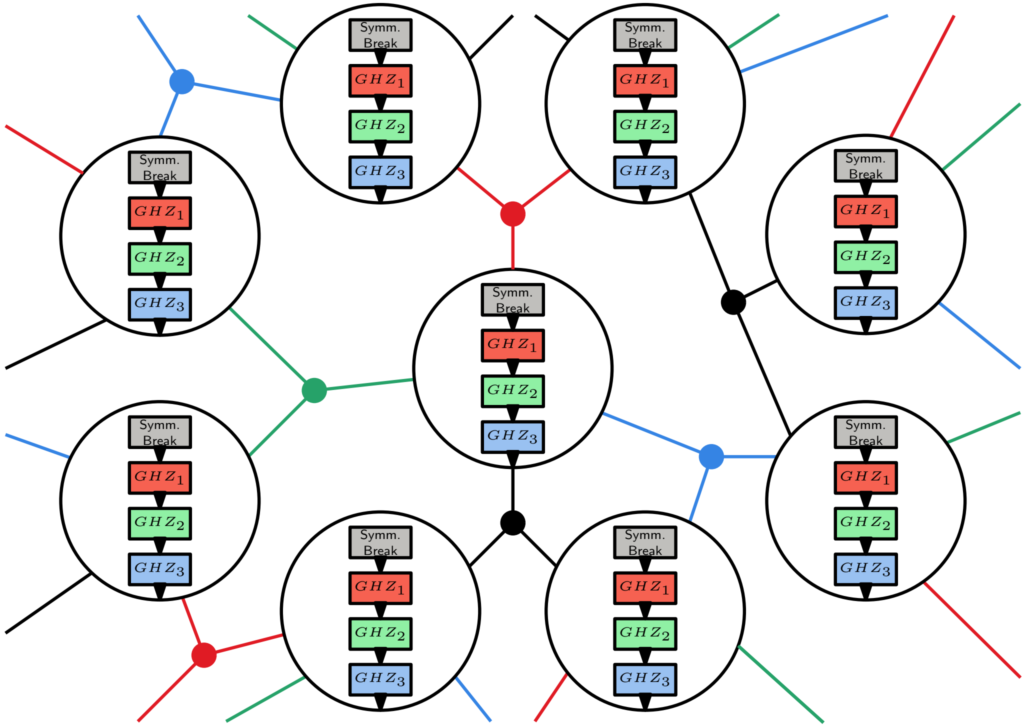

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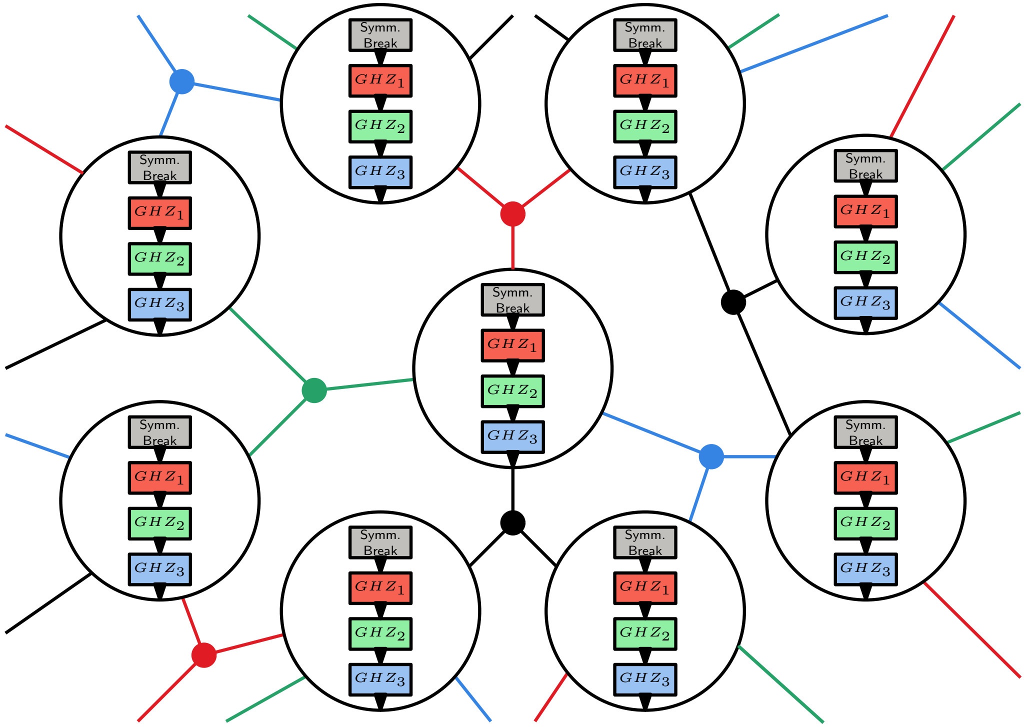

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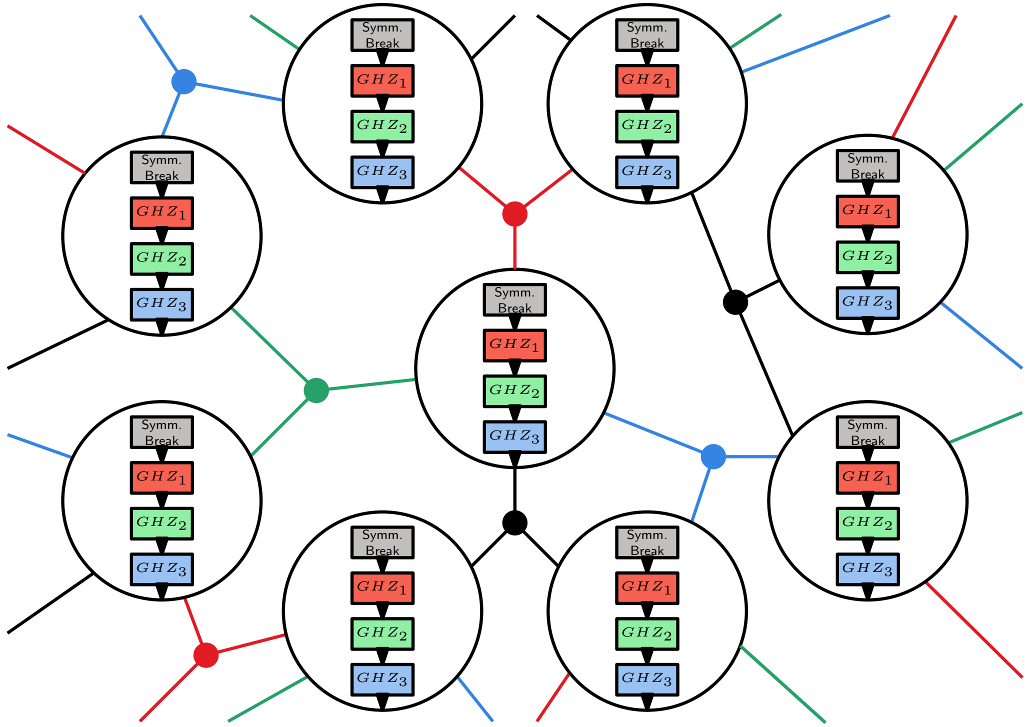

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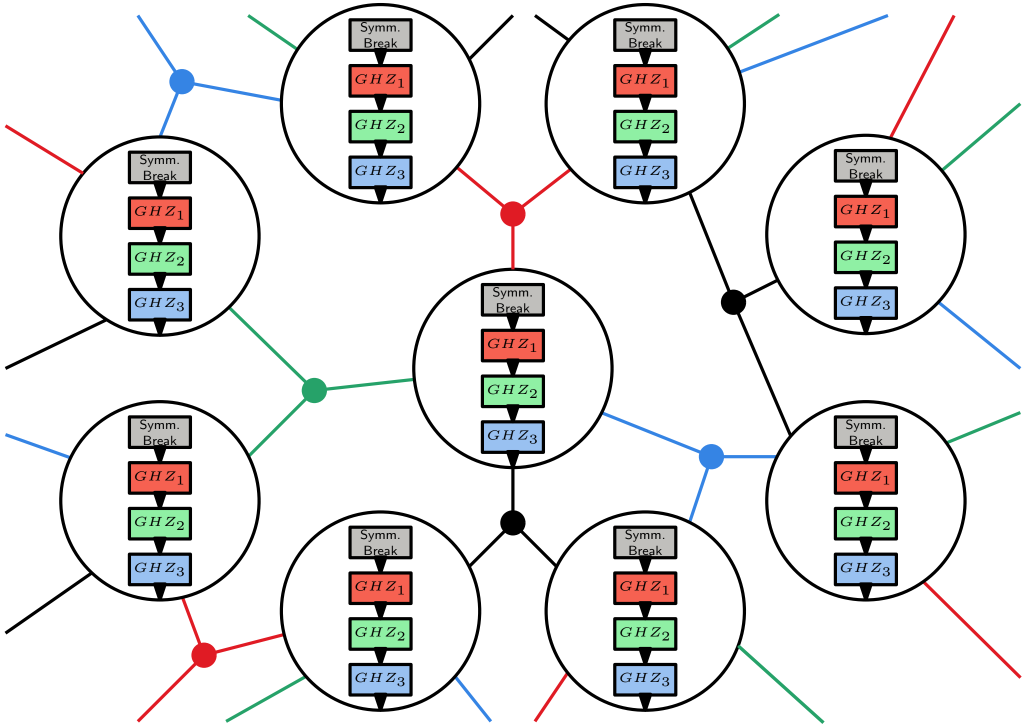

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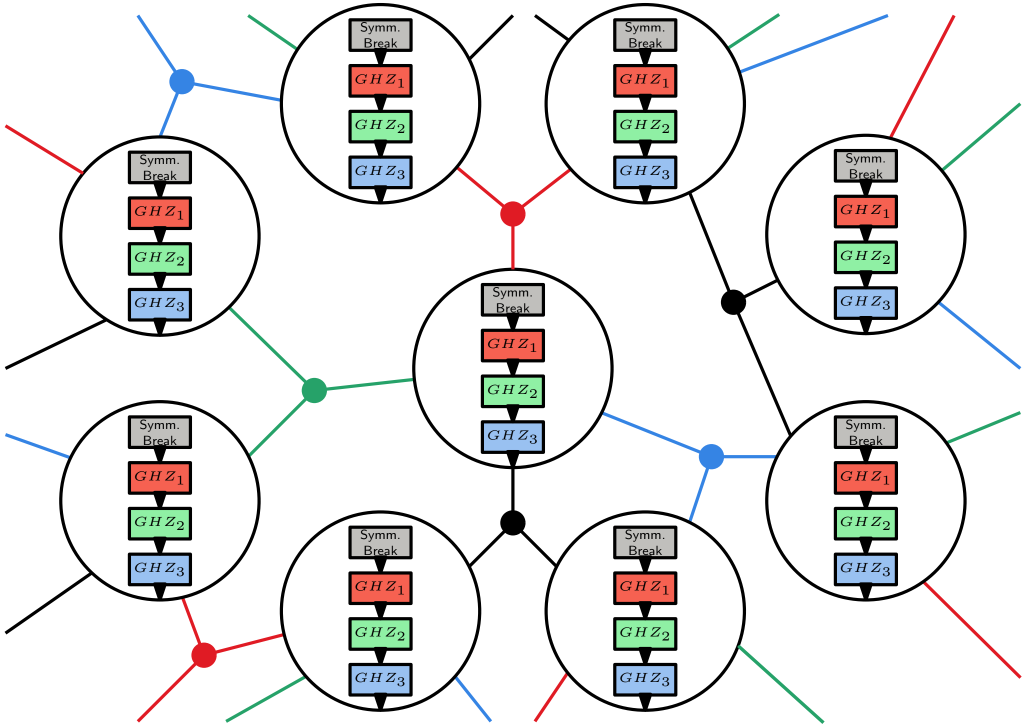

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- ...
- solve GHZ_Δ game with the last neigh.

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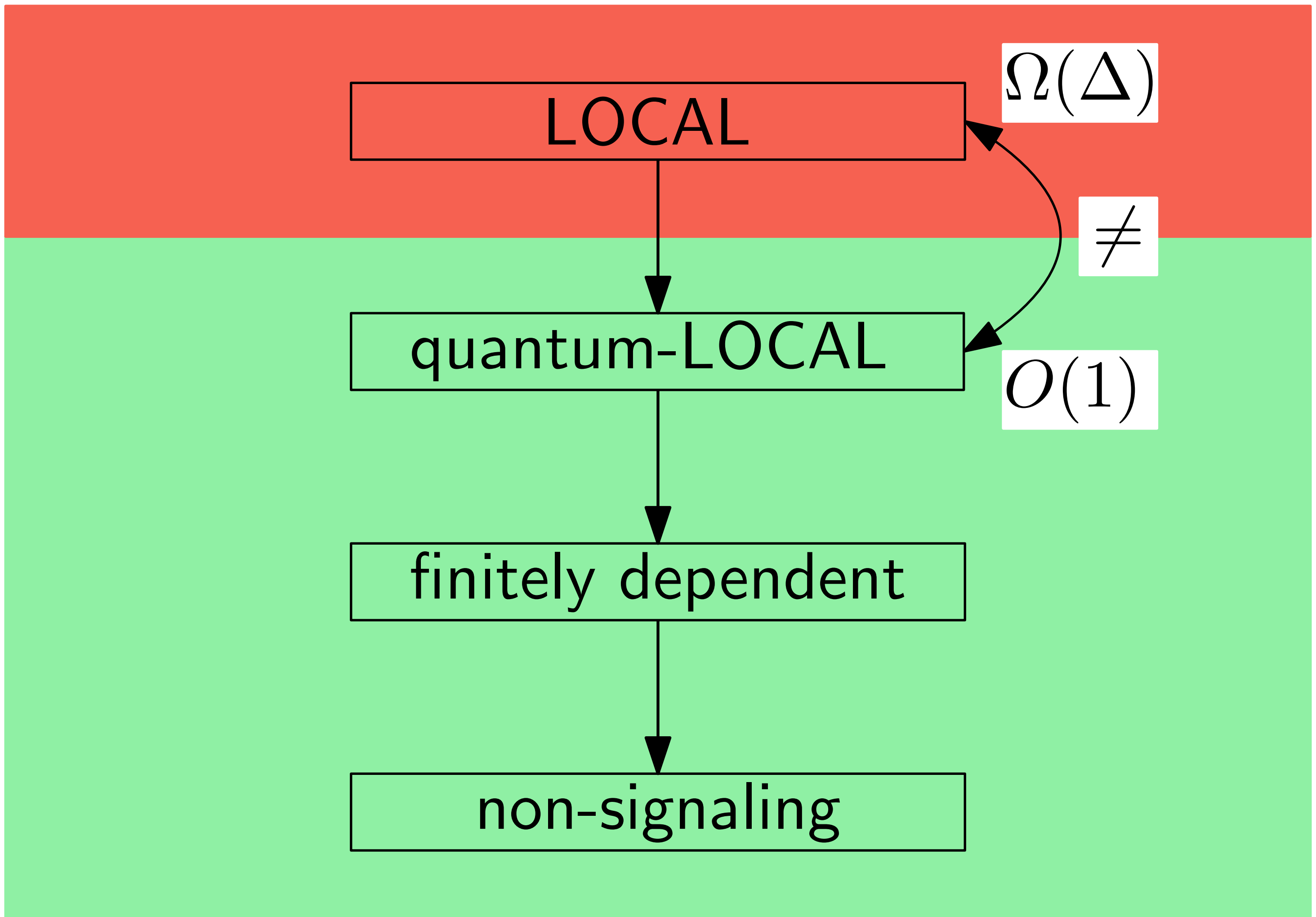

Summary

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Linear Progams (also SDPs) : no separation

New Limits on Distributed Quantum Advantage: Dequantizing Linear Programs

A. Balliu, C. Coupette, A. Cruciani, F. D'Amore, M. Equi, H. Lievonen, A. Modanese, D. Olivetti, J. Suomela

DISC 2025

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LCLs : $O(\Delta)$ LOCAL vs $O(1)$ quantum-LOCAL

Distributed Quantum Advantage for Local Problems

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- only one previous result for LOCAL, and not for LCLs
- most previous literature focused on other models like CONGEST

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New Limits on Distributed Quantum Advantage: Dequantizing Linear Programs
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DISC 2025

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Quantum Pattern Matching in Generalised Degenerate Strings
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Thank you! Questions?